PA1:

$l_{2AB} = 5m$ $l_{BH} = 2m$ $l_{CD} = 6$ $\varphi_2 = 10°$

$l_{3K} = 9,5m$ $l_{HE} = 1m$ $l_{AD} = 3$ $\omega_2 = 5$ (rad/s): vận tốc góc khâu 2 so với khâu cố định

$\varepsilon_2 : 10 (rad/s^2)$ vận tốc góc so với khâu cố định

$P_3 = 210N$ $M_3 = 30Nm$ $m_3 = 2kg$ $m_4 = 1kg$.

$\alpha = 10°$ $N_4 = 150Nm$ $JS_3 = 2,5 \cdot kg\,m/s^2$

Câu 1:

Số khâu động: $n = 3$ (2, 3, 4)

Số khớp thấp: $P_t = 4$ (A, B, C, D).

Số khớp cao: $P_c = 0$

$W = 3.n - (2.P_t + P_c.1) = 3.3 - (2.4 + 0.1) = 1.$

Chọn khâu 2 và khớp A làm khâu dẫn $\Rightarrow$ còn lại 2 khâu động
3 khớp thấp

$\Rightarrow$ Có cấu loại 2.

Câu 2:

$v_B = \omega_2 . l_{AB} = 5 . 5 = 25\ m/s.$

$\vec{v_b} = \vec{v_A} + \vec{v_{BA}}$

$\vec{v_A} = 0$

$\vec{v_{BA}}$: $\perp AB$, theo chiều quay ω_2, 25 m/s.

$\mu_v = \frac{v_b}{P_b} = \frac{25}{2,5} = 10 \frac{(m/s)}{cm}$.

$\vec{v_C} = \vec{v_B}$ + $\vec{v_{CB}}$ = $\vec{v_D}$ + $\vec{v_{CD}}$

$\vec{v_B}$: $\perp AB$, theo chiều quay ω_2, $25 m/s$

$\vec{v_{CB}}$: $\perp BC$, theo chiều quay ω_3, $\omega_3 . l_{BC}$

$\vec{v_D} = 0$

$\vec{v_{CD}}$: $\perp CD$, theo chiều quay ω_4, $\omega_4 . l_{CD}$

$v_C = P_C . \mu_v = 8,18 . 10 = 81,8 m/s$.

$\Rightarrow \omega_4 . l_{CD} = 81,8 \Rightarrow \omega_4 = \frac{81,8}{6} = 15,6 rad/s$.

$|v_{CB}| = 6,9$

$\Rightarrow v_{CB} = 6,9 . 10 = 69 m/s$.

$\Rightarrow \omega_3 . l_{BC} = 69 \Rightarrow \omega_3 = \frac{69}{4,5} = 15,3 rad/s$.

$v_H = P_M . P_H . \mu_v = 4,63 . 10 = 46,3 m/s$.

$|v_E| = 5,56 \Rightarrow v_E = P_E . 10 = 55,6 m/s$.

$a_n = \frac{125}{} \frac{m/s^2}{}$

Date . . . No.

Câu 5:

$\vec{a_B} = \vec{a^n_{BA}} + \vec{a^t_{BA}}$

I { $\perp BA$; $B \to A$; $W_2^2 . l_{AB}$ } { $\perp BA$; theo c quay ε_2 ; $l_{AB} . \varepsilon_2$ }

$= 5^2 . 5$ $= 5 . 10$

$= 125 \ m/s^2$ $= 50 \ (m/s^2)$

$\Rightarrow a_B = \sqrt{125^2 + 50^2} \approx 139,63 \ m/s^2$

$\Rightarrow \pi_a = \frac{a_B}{|\pi_B|} = \frac{139,63}{|\pi_B|} = 139,63 \ \frac{m/s^2}{cm}$

$\vec{a_C} = \vec{a_B} + \vec{a^n_{BC}} + \vec{a^t_{BC}} = \vec{a_{CD}} + \vec{a^t_{CD}}$

$\pi_a = 125 \left| \frac{m/s^2}{cm} \right|$.

$|a^n_{BA}| = \frac{125}{125} = 1 \ (cm)$

$|a^t_{BA}| = \frac{50}{125} = 0,4 \ (cm)$.

$\Rightarrow a_B = \sqrt{125^2 + 50^2} \approx 134,63 \ m/s^2$

$\vec{a_C} = \vec{a_B} + \vec{a^n_{CB}} + \vec{a^t_{CB}} = \vec{a^n_{CD}} + \vec{a^t_{CD}}$

{ $\perp$; $\perp$; $\perp$ } { $\parallel BC$; $C \to B$; $W_3^2 . l_{CB}$ } { $\perp BC$; theo c quay ε_3 ; $l_{BC} . \varepsilon_3$ } { $\parallel CD$; $C \to D$; $W_4^2 . l_{CD}$ } { $\perp CD$; theo c quay ε_4 ; $l_{CD} . \varepsilon_4$ }

$= 1053,905 \ m/s^2$ $= 1363^2 . 6$

$\Rightarrow |a^n_{CB}| = 8,93$ $= 1119,7 . m/s^2$

$\Rightarrow |a^n_{CD}| = 8,92$

$a_C = \pi_C . \pi_a = 9,23 . 125 = 1153,75 \ (m/s^2)$

$|a^t_{CB}| = 0,48 \Rightarrow a^t_{CB} = 0,48 . 125 = 60 \ (m/s^2) \Rightarrow \varepsilon_3 = \frac{60}{9,5} \approx 13,33 \ rad/s^2$

$|a^t_{CD}| = 2,3 \Rightarrow a^t_{CD} = 2,3 . 125 = 287,5 \ (m/s^2) \Rightarrow \varepsilon_4 = \frac{287,5}{6} = 47,9 \ rad/s^2$

$a_M = \pi_M . \pi_a = 9,95 . 125 = 556,25 \ m/s^2$.

$a_E = \pi_E . \pi_a = 5,2 . 125 = 650 \ m/s^2$

HONG HA